

Tony Kipchirchir

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Professional Summary

Electrical and Electronics Engineering graduate combining formal academic training with over four years of hands-on PCB repair and hardware troubleshooting experience. Capable of diagnosing complex multi-layer boards down to the component level, alongside designing custom embedded prototypes like a real-time Visible Light Communication (VLC) transceiver. Seeking to apply practical circuit debugging, RF signal testing, and embedded programming skills to an entry-level engineering role in hardware development, telecommunications, or signal processing.

Skills

Technical Skills:

- **Hardware & Diagnostics:** Component-Level Troubleshooting, SMD Micro-soldering, Schematic Tracing, PCB Rework, Multimeters, Oscilloscopes, Spectrum Analyzers
- **Embedded & Software:** C, C++, Python, Arduino, MATLAB, Simulink, TINA TI
- **Protocols & Signals:** I2C, SPI, UART, PWM, ADC, RF Diagnostics, VSWR Testing

Leadership & Interpersonal:

- Project Mentorship, Technical Instruction, Client Relations, Independent Project Management, Workflow Optimization

Work Experience

Electronics Repair Specialist | Dr. Cellulaire | Machakos, Kenya | Jan 2021 – Present

- Restored 1000+ non-functional devices using component-level diagnostics, schematic tracing, and precision SMD micro-soldering.
- Streamlined repair workflows and managed inventory, achieving high customer satisfaction across independent service operations.

STEM Educator & KSEF Project Advisor | Lukenya Boys High School | Feb 2026 – Mar 2026

- Directed hardware engineering initiatives, guiding students to successfully build and test a Minimum Viable Product.
- Mentored teams in rapid prototyping and iterative design, delivering functional projects under strict exhibition deadlines.

Industrial Attaché | Kenya Civil Aviation Authority (KCAA) | Mombasa, Kenya | Jun 2025 – Aug 2025

- Executed RF signal integrity and VSWR assessments using spectrum analyzers to ensure optimal transmission quality.
- Collaborated with senior engineers to isolate transmission faults, successfully minimizing critical communication downtime.

Arduino Lead | Techno Engineering Association | Machakos, Kenya | Jan 2024 – Dec 2024

- Mentored 50+ engineering students in microcontroller debugging, bridging circuit theory with functional sensor integration.
- Led practical sessions on embedded C programming, hardware prototyping, and sensor deployment for university peers.

Technical Attaché | Carl Care Service Center | Kitengela, Kenya | Jun 2023 – Aug 2023

- Diagnosed consumer electronics using manufacturer schematics and multimeters, adhering to strict quality assurance standards.
- Documented recurring hardware faults, contributing directly to improved diagnostic workflow efficiency for the team.

Domestic Electrical Installer | Freelance | Dec 2019 – Aug 2020

- Managed comprehensive domestic electrical installations, ensuring 100% compliance with standard safety regulations.

Projects

Visible Light Communication (VLC) Transceiver | Feb 2026 – Present

- Designed and built a real-time VLC transceiver for wireless optical data transmission via high-speed LED modulation.
- Developed an analog optical receiver front-end (BPW34 photodiode, OPA380 amplifier) for MHz-range light signal conditioning.

Underground Cable Fault Detection System | Jan 2025 – Mar 2025

- Prototyped a portable diagnostic tool using Time-Domain Reflectometry principles to precisely pinpoint faults in buried infrastructure.
- Programmed a microcontroller to calculate distance-to-fault metrics from voltage drops, reducing physical troubleshooting labour.

IoT-Based Secure Access & Data Logging System | Aug 2024 – Sep 2024

- Engineered a hardware access control prototype integrating serial sensor inputs with real-time cloud-based monitoring.
- Interfaced RFID authentication with a Wi-Fi microcontroller for encrypted database logging and traceable personnel telemetry.

Education

Machakos University | Bachelor of Science - Electrical and Electronics Engineering